Contents

PART-I : PARTIAL DIFFERENTIAL EQUATION

PART-II : ORDINARY DIFFERENTIAL EQUATION

PART-III : STATICS

PART 2

ORDINARY DIFFERENTIAL EQUATION

Topics:

1. Formulation.
2. Equations of 1^{st} order and 1^{st} degree.
3. Trajectories.
4. Equations of 1st order but not of 1^{st} degree.
5. Linear differential equations with const. coefficient.
6. Homogenous linear equations or cauchy euler equation.
7. Method of variation of parameters.
8. Linear equations of second order.
9. Laplace and Inverse and its application to IVPs.

DIFFERENTIAL EQUATIONS : THEIR FORMATION AND SOLUTIONS

Differential Equation - An equation involving one (or more) dependent variable and its derivatives with respect to one or more independent variables is called a differential equation.

e.g. $\frac{dy}{dx} = \cos x, \quad y^2 \frac{\partial z}{\partial x} + xy \frac{\partial z}{\partial y} = nzx.$

Ordinary : Single independent variable.

e.g. $g\left(x, y(x), \frac{dy}{dx}, \frac{d^2y}{dx^2}, \ldots\ldots, \frac{d^n y}{dx^n}\right) = 0$

Partial : DE involves two or more independent variables.

e.g. $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0$

$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} - z = 0$

Order : The order of a differential equation is the order of the highest order derivative appearing in the equation.

e.g. $\frac{d^2y}{dx^2} + y = x^2$ is second order. It is a positive integer.

$\frac{d^4y}{dx^4} = 0$ is fourth order differential equation.

Degree : The degree of a differential equation is the highest exponent of the highest order derivative appearing in it after the equation has been expressed in the form free from radicals and any fractional power of the derivatives or negative Power.

e.g. $y - x\frac{dy}{dx} = r\sqrt{1 + \left(\frac{dy}{dx}\right)^3}$ is of degree 3.

Similarly, $y = x\frac{dy}{dx} + \frac{a}{dy/dx}$ is of degree two.

CLASSIFICATION OF DIFFERENTIAL EQUATION

Depending upon the degree of dependent variables and its derivatives.

Linear   Non-Linear

Dependent variable and its derivatives occur to the first degree only and not as higher power or products.

Solution:

- → General Solution.
- → Particular Solution.
- → Singular Solution.

Formulation of DE

Given a family of curves containing n arbitrary constants.

Working Rule:

1. Differentiat the equation of family of curves, n times so as to get n additional equations containing the n arbitrary consts and derivatives.
2. Eliminate 'n' arbitrary const from the $(n+1)$ equations. Thus we obtain the required DE involving a derivative of n^{th} order.

Q. Find the differential equation of all circles of radius 'a'.

WRONSKIAN

The wronskian of n function $y_1(x), y_2(x)....y_n(x)$ is denoted by $W(x)$ or $W(y_1, y_2...y_n)(x)$ and is denoted by the determinant.

$$W(y_1, y_2, ...y_n)(x) = \begin{vmatrix} y_1 & y_2 & ...y_n \\ y_1' & y_2' & ...y_n' \\ \downarrow & & \\ y_1^{(n-1)'} & y_2^{(n-1)'} & ...y_n^{(n-1)'} \end{vmatrix}$$

Note

- n functions $y_1(x), y_2(x)----y_n(x)$ are linearly dependent if there exist constants $c_1, c_2 .....c_n$ (not all zero) such that $c_1 y_1 + c_2 y_2 + ... + c_n y_n = 0$. If $c_1 = c_2 ..... = c_n = 0$ *then* $y_1, y_2 .... y_n$ are said to be linearly independent.
- Given two functions $f(x)$ and $g(x)$ that are differentiable on some interval I, then,

1. If $W(f, g)(x_0) \neq$ for some x_0 in I, then $f(x)$ and $g(x)$ are linearly independent on the interval I.
2. If $f(x), g(x)$ are linearly dependent on I then $W(f, g)(x) = 0$ for all x in the interval I.
3. When $f(x)$ *and* $g(x)$ are the two solutions of the DE, then they are linearly dependent if and only if their wronkskian is identically zero.

Q. Prove that the function 1, x, x^2 are linearly independent. Hence form the differential equation whose solution are 1, x, x^2.

Differential Equations arising from physical situations.

(1) **Population Model** - Let $N(t)$ denote population at time t. we assume that rate of change of N at time 't' is proportional to population $N(t)$, present at the time t, then,

$$\frac{d}{dt}N(t) \propto N(t)$$

(2) **Newton's Law of cooling** - Temperature variations of hot object kept in a surrounding which is kept at a const. temperature T_0.

"Newtons' law of cooling states that the rate at which temperature $T(t)$ changes in a cooling body is proportional to the difference between the temperature of the body and the constant temperature T_0 of the surrounding medium.

$$\frac{d}{dt}T(t) \propto T(t) - T_0$$

(3) **Radioactive Delay** - In physics it has been noticed that the radioactive material ,at time t, decays at rate proportional to its amount' $y(t)$.

$$\frac{d}{dt}(y(t)) = k\, y(t)$$

Equations of First Order And First Degree

Two standard forms

$$\frac{dy}{dx} = f(x, y) \qquad Mdx + Ndy = 0$$

Method for Solving

(1) Separation of variables or Reducible to SoV.

(2) Homogeneous equations or Reducible to Hom.

(3) Exact differential equation.

(4) Linear form or reducible to linear form.

(5) Integrating factors, when $Mdx + Ndy = 0$ is not exact.

(1) Separation of Variables

Step 1 : Let $\frac{dy}{dx} = f_1(x)f_2(y)$

Step 2 : Let $\frac{1}{f_2(y)}dy = f_1(x)dx$

Step 3 : Integrating both sides

$$\int \frac{L}{f_2(y)}dy = \int f_1(x)dx + c$$

C: constant of integration is chosen in a way so as to get the final solution in simplest form.

e.g. c may be $\log c$, $\tan^{-1}c$, e^c etc.

Q. $\frac{dy}{dx} = e^{x-y} + x^2e^{-y}$

Ans: $e^y = \frac{x^3}{3} + e^x + c$

Q. $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$

Ans: $e^y = \frac{x^3}{3} + e^x + c$

Q. Solve $\left(\frac{dy}{dx}\right)\tan y = \sin(x+y) + \sin(x-y)$

Ans: $\sec y = -2\cos x + c$

Note :

Equations of the form : $dy/dx = f(ax+by+c)$ or $dy/dx = f(ax+by)$ can be reduced to an equation in which variables can be separated.

put $ax+by+c = v$ or $ax+by = v$

Solve $dy/dx = (4x+6y+5)/(3y+2x+4)$

Ans: $y - 2x + (3/8)\log(16x+24y+23) = c'$

Homogenous Equation

Step-1: Write the equation in $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$ form.

Step-2: Put $y = Vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

Step-3: Separate the variables x and v.

Q. Solve $xdy - ydx = (x^2 + y^2)^{1/2} dx$

Ans: $x^2 c = y + \sqrt{x^2 + y^2}$,

Note: Equations of the form

$$\frac{dy}{dx} = \frac{ax + by + c}{a'x + b'y + c'} \text{ where } \frac{a}{a'} \neq \frac{b}{b'}$$

can be reduced to homogenous form.

take $x = X + h$ and $y = Y + k$

$dx = dX$ $dy = dY$

Thus, $$\frac{dy}{dx} = \frac{dY}{dX} = \frac{a(X+h) + b(Y+k) + c}{a'(X+h) + b'(Y+k) + c'}$$

$$= \frac{aX + bY + (ah + bk + c)}{a'X + b'Y + (a'h + b'k + c')}$$

Now, solve for 'h' and 'k'

$$\begin{cases} ah + bk + c = 0 \\ a'h + b'k + c' = 0 \end{cases}$$

$$\frac{dy}{dx} = \frac{aX + bY}{a'X + b'(Y)} = \frac{a + b\left(\frac{Y}{X}\right)}{a' + b'\left(\frac{Y}{X}\right)}$$

Exact Differential Equation

Let M and N be functions of x, y then the equation $Mdx+Ndy=0$ is called Exact, when there exist a function $f(x, y)$ such that.

$$Mdx + Ndy = df$$

The necessary and sufficient condition for the differential equation $Mdx + Ndy = 0$ to be exact is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Working Rule: Given $Mdx + Ndy = 0$ and $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$ (i.e. exact)

(1) Integrate M w.r.t. x treating y as a const.

(2) Integrate w.r.t. y only those terms of N which do not contain x.

(3) Equate the sum of these two integrats to an arbitrary const. and thus we obtain the required solution.

$$\int Mdx + \int (\text{Terms in } N \text{ not containing } x)\, dy = C$$

[treating y as const.]

Q. Solve: $\left(x^2 - 4xy - 2y^2\right)dx + \left(y^2 - 4xy - 2x^2\right)dy = 0$

INTEGRATING FACTOR

If $Mdx + Ndy = 0$ is not exact it can always be made exact by multiplying by some function of x and y. Such a multiplier is called Integrating Factor I.F.

Rules For Finding I.F

(1) By inspection : Rearrange the terms.

(2) $Mdx + Ndy = 0$: Homogenous and $Mx + Ny \neq 0$ then, I.F $= \dfrac{1}{Mx + Ny}$

(3) $Mdx + Ndy = 0$ is of the form.

$f_1(xy)ydx + f_2(xy)xdy = 0$ and $Mx - Ny \neq 0$ then , I.F $= \dfrac{1}{Mx - Ny}$

(4) If $\dfrac{1}{N}\left(\dfrac{\partial M}{\partial y} - \dfrac{\partial N}{\partial x}\right) = f(x)$, then $I.F. = e^{\int f(x)dx}$

(5) If $\dfrac{1}{M}\left(\dfrac{\partial N}{\partial x} - \dfrac{\partial M}{\partial y}\right) = f(y)$ I.F $= e^{\int f(y)dy}$

(6) $Mdx + Ndy = 0$ is of the form

$x^{\alpha} y^{\beta} (mydx + nxdy) + x^{\alpha'} y^{\beta'} (m'ydx + n'xdy) = 0$

I.F. $x^h y^k$,where h, k are obtained by applying the condition that the given equation must become exact after multiplying by $x^h.y^k$

Solve: $(x^3 + xy^2 + a^2y)dx + (y^3 + yx^2 - a^2x)dy = 0$

Solve: $(x^2y - 2xy^2)\,dx - (x^3 - 3x^2y)\,dy = 0$

Solve: $(xy\sin xy + \cos xy)\,y\,dx + (xy\sin xy - \cos xy)\,x\,dy = 0$

Solve: $\left(y+\frac{y^3}{3}+\frac{x^2}{2}\right)dx+\left(\frac{1}{4}\right)(x+xy^2)dy=0$

Solve: $(y^2+2x^2y)dx+(2x^3-xy)dy=0$

Linear Differential Equation

$$\frac{dy}{dx} + Py = Q$$

$$\text{I.F.} = e^{\int P dx}$$

Solution $y \times \text{I.F.} = \int [Q \times \text{I.F.}] dx + c$

Note: $$\frac{dx}{dy} + p(y)x = Q(y)$$

$$\text{I.F.} = e^{\int P dy}$$

Sol. $x \times \text{I.F.} = \int [Q \times \text{I.F.}] dy + c$

Q. Solve $x\cos x\left(\frac{dy}{dx}\right)+y(x\sin x+\cos x)=1$

Note:

(1) Equation reducible to linear form.

$$f'(y)\frac{dy}{dx}+P.f(y) = Q$$

Put $f(y) = v$

(2) Bernoullie's equation: $\frac{dy}{dx}+Py = Qy^n$

Divide by y^n

$$y^{-n}\left(\frac{dy}{dx}\right)+Py^{1-n} = Q$$

Put $y^{1-n} = v$

Q. $\dfrac{dy}{dx} = e^{x-y}(e^x - e^y)$

Trajectories

"A curve which cuts every member of a given family of curve in accordance with some given law is called trajectory of the given family of curves"

Orthogonal Trajectory: If a curve cuts every member of a given family of curves at right angles, it is called an orthogonal trajectory.

e.g. Circle: $x^2 + y^2 = a^2$, a, m are parameters.
$y = mx$

Determination of orthogonal Trajectories in Cartesian coordinates.

Let given family of curves : $f(x, y, c) = 0$ (c is a parameter)

↓

Find its differential equation.

↓

Replace dy/dx by $-dx/dy$.

↓

Solve the new DE. This is the required orthogonal trajectory.

Q. Find the orthogonal trajectories of the family of coaxial circles $x^2 + y^2 + 2gx + c = 0$, where g is the parameter.

Orthogonal Tranjectories in Polar Coordinates.

Let given family be $f(r, \theta, c) = 0$

$\downarrow$

Find DE

$\downarrow$

Replace $\frac{dr}{d\theta} = -r^2 \frac{d\theta}{dr}$

$\downarrow$

Solve. This is the required orthogonal trajectory.

Q. Find the orthogonal trajectory of the Cardials $r = a(1 - \cos\theta)$,'a' being parameter.

Ans. $r = b(1 + \cos\theta)$

Oblique Trajectories

Let given family of curve $f(x,y,c)=0$

$\downarrow$

Find DE.

$\downarrow$

Replace $\frac{dy}{dx}$ by $\left[\frac{dy/dx+\tan\alpha}{1-(dy/dx)\tan\alpha}\right]$

$\downarrow$

Solve

Q. Find the family of curves whose tangents form the angle of $\frac{\pi}{4}$ with the hyperbola $xy = c$.

Equations of the first order but not of the first degree.

Denote $\frac{dy}{dx} = p$

The equation of the form

$P_o p_n + P_1 p^{n-1} + \ldots\ldots + P_n = 0$ where $P_0, P_1 - P_n$ are functions of the x and y

Two forms of the solution

(1) Cartesian Form : $y^2 = 2cx + c^2$

(2) Parametric Form: $x = f_1(p,c)$
$y = f_2(p,c)$

Method of Solving .

(1) Solvable for p

(2) Solvable for x

(3) Solvable for y

(4) Clairaut's form.

(1) Solvable for p

Form: $[p - f_1(x,y)][p - f_2(x,y)....[p - f_n(x,y) = 0$

- Equating each factor to zero and solve these n factors .
- Let solution be $f_1(x,y,c_1) = 0$, $f_2(x,y,c_2) = 0$, $f_n(x,y,c_n) = 0$
- Combine $f_1(x,y,c)\, f_2(x,y,c)\_\_ f_n(x,y,c) = 0$

Note: General solution can have only one arbitrary const.

Q. $x^2p^2 + xyp - 6y^2 = 0$

(2) **Equations solvable for x**

$$x = f(y,p)$$

$$\downarrow$$

Differentiate w.r.t., y and replace $\frac{dx}{dy} \to \frac{1}{p}$

$$\downarrow$$

$$\frac{1}{p} = \phi\left(y, p, \frac{dp}{dy}\right)$$

$$\downarrow$$

This is a equation involving two variables y and p

$\therefore$ Solve to get $\psi(y,p,c) = 0$

$$\downarrow$$

Eliminate p between $x = f(y, p)$

and $\psi(y, p, c) = 0$

This is the solution

Note :

(1) If it is not possible to eliminate p, then we solve $x = f(y, p)$ and $\psi(y, p, c) = 0$ to express x and y in terms of 'p' and 'c' i.e $x = f_1(p,c)$ and $y = f_2(p,c)$

(2) If $\frac{1}{p} = \phi\left(y, p, \frac{dp}{dy}\right)$ can be expressed as $\phi_1(y,p).\phi_2\left(y, p, \frac{dp}{dy}\right) = 0$ then ignore $\phi_1(y,p)$ which doesn't involve $\frac{dp}{dy}$. Actually eliminating p between $x = f(y,p)$ and $\phi_1(y,p)$ will give singular solution of $x = f(y,p)$

Q. $y = 2px + y^2p^3$

Q. $x = py + p^2$

$$y = (c + \sin^{-1} p)(1 - p^2)^{-1/2} - p$$
$$x = p(c + \sin^{-1} p)(1 - p^2)^{-1/2}$$

Equations solvable for y

$$y = F(x,p) \qquad \text{...(1)}$$

$$\downarrow$$

Differentiate w.r.t x, write p for $\frac{dy}{dx}$ this equation will be of the form $p=\phi\left(x,p,\frac{dp}{dx}\right)$, solve to get

$$\psi(x,p,c)=0 \qquad \text{...(3)}$$

$$\downarrow$$

Eliminate p between $y = F(x, p)$ and $\psi(x, p, c)$

'or'

If it is not possible to eliminate p between (1) and (3), express x and y in terms of p and c.

$$x = f_1(p,c) \quad and \quad y = f_2(p,c)$$

Note

(1) If $p=\phi\left(x, p, \frac{dp}{dx}\right)$ can be expressed as $\phi_1(x,p)\phi_2\left(x,p,{}^{dp}/_{dx}\right)=0$, ignore first factor.

(2) Singular Solution : Eliminate p between $\phi_1(x,p)$ and $y = F(x, p)$.

(3) $y = 3x + \log p$

Ans. $y=3x+\log\left\{\frac{3}{1-ce^{3x}}\right\}$

Equations in Clairaut's Form

Form: $y = px + f(p)$

Its general solution is given by $y = cx + f(c)$

Q. $p = \log(px - y)$

Equations Reducible to Clairaut's form

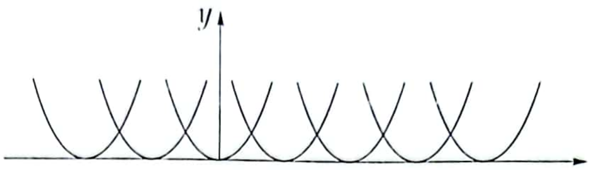
By suitable substitutions, equations can be reduced to Clairaut's form.

Q. To solve $y^2 = \left(\frac{py}{x}\right)x^2 + f\left(\frac{py}{x}\right)$

Singular Solutions

Consider $(y')^2 - 4y = 0$

Its general Solution is $y = (x+c)^2$



Besides these curves , $y = 0$ also satisfies the given D.E. However, this function is not contained in the general solution ! such solutions are called singular solutions.

Envelope and Singular Solution

Whenever the family of curves $\phi(x,y,c)=0$ represented by the differential equation $f(x,y,p)=0$ possesses an envelope the equation of the envelope is the singular solution of the differential equation.

p discriminant and c discriminant.

p discriminant

Let the given equation be $f(x,y,p)=0$

The relation obtained by eliminating "p" between $f(x,y,p)=0$ and $\partial f/\partial p=0$ is p-discriminant.

c-discriminate.

Let $\phi(x,y,c)$ be the general solution of $f(x,y,p)$.

The relation obtained by eliminating 'c' between $\phi(x,y,c)$ and $\frac{\partial\phi}{\partial c}=0$ is c-discriminant.

General Algorithm of Finding Singular Points/Solution

- We make use of p-discriminant and c-discriminant.
- Find the equations of the p-discriminant and c-discriminant.
  - → $\psi_p(x,y)=0$: equation of p-discriminant.
  - → $\psi_c(x,y)=0$: equation of c-discriminant.

In general,

The equation of p-discriminant can be factored into the product of three function :

$$\psi_p(x,y)=E*T^2*C=0$$

Where E = Envelope

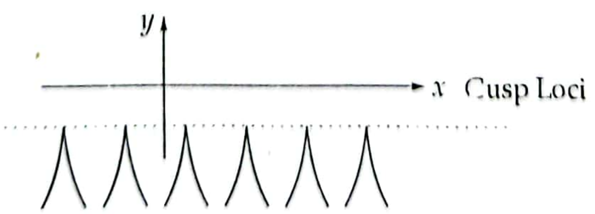
T = Tac locus

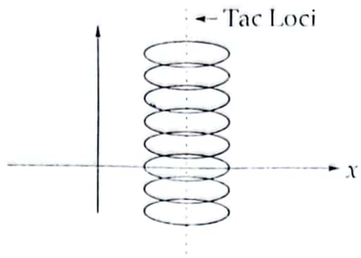
C = Cusp locus

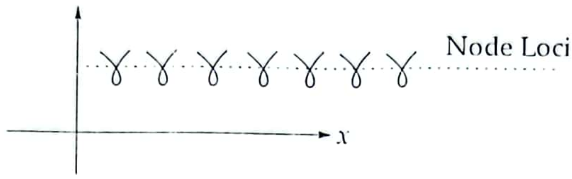
Similarly, the equation of c-discriminant can be factored into the product of three functions :

$$\psi_c(x,y)=E*N^2*C^3=0$$

E = Envelope, N = Node locus, C : Cusp locus, these TNC will not satisfy the D.E. Clear from the graphical representation.







TNC are called Extraneously points Don't satisfy the D.E Only Envelope Satisfies DE, and is consider singular solution. Since the envelope is present in both the discriminants as a first degree factor, this allows to find the equation of the envelope.

Q. Find general and singular solution of $9p^2(2-y)^2=4(3-y)$.

Ans. $(x + c)^2 = y^2 (3 - y)$ $y = 3$ is G.S.

Linear Differential Equations with Constant Coefficients

General Form:

$$\frac{d^n y}{dx^n}+a_1\frac{d^{n-1}y}{dx^{n-1}}+a_2\frac{d^{n-2}y}{dx^{n-2}}+\ldots+a_n y = X$$

where X is a function of x only and $a_1, a_2 \ldots a_n$ const.

Use Symbols $D^n : \frac{d^n}{dx^n}$, $D^{n-1} : \frac{d^{n-1}}{dx^{n-1}} \ldots$

$$f(D)y = X \qquad \ldots(1)$$

where $f(D) = D^n + a_1 D^{n-1} + \ldots + a_n$. The general solution of (1) is given by

$$y = \text{C.F.} \quad + \quad \text{P.I} \quad = \quad y_c + y_p$$

Complementary function   Particular Integral

$$f(D)y = 0$$

Finding Complementary Function (C.E)

$$f(D)y = 0$$

$$\downarrow$$

Replace D by m to get Auxiliary equation $f(m) = 0 \rightarrow m_1, m_2 \ldots$ general n roots.

Case-I: All roots of $f(m)=0$ are real & distinct $m_1, m_2 \ldots m_n$

$$y_c = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \ldots + c_n e^{m_n x}$$

Case-II: Equal Roots.

$m_1, m_1, m_3 \ldots, m_n$.

$$y_c = (c_1 + c_2 x)e^{m_1 x} + c_3 e^{m_3 x} + \ldots + c_n e^{m_n x}$$

Case-III: AE has complex roots

$$m_1 = \alpha + i\beta,\ m_2 = \alpha - i\beta,\ m_3, m_4 \ldots m_n$$

$$y_c = e^{\alpha x}(c_1 \cos\beta x + c_2 \sin\beta x) + c_3 e^{m_3 x} + \ldots$$

Let's say m_1, m_2 are repeated twice.

Replace $c_1 \to c_1 + c_2 x$
$c_2 \to c_3 + c_4 x$

Case IV: AE has surd roots. $\left(\alpha \pm \sqrt{\beta}\right)$

$e^{\alpha x}(c_1 \cosh\sqrt{\beta} + c_2 \sinh\sqrt{\beta})$ **Rest terms : as usual.**

Q. Solve $\dfrac{d^3y}{dx^3} + 6\dfrac{d^2y}{dx^2} + 11\dfrac{dy}{dx} + 6y = 0$

Q. $(D^4 - 81)y = 0$

Q. $(D^2 + 1)^2 (D^2 + D + 1)y = 0$

Determination of Particular Integral

$\frac{1}{f(D)}X$ is inverse of operator $f(D)$

→ PI will never contain arbitrary const.

General Method of Finding PI

$$\frac{1}{D-\alpha}X = e^{\alpha x}\int e^{-\alpha x}X\,dx$$

or

$$\frac{1}{D+\alpha}X = e^{-\alpha x}\int e^{\alpha x}X\,dx$$

Working Rule

M-I: The Operator $1/f(D)$ may be factored into linear factors :

Then P.I $= \frac{1}{D-\alpha_1}\frac{1}{D-\alpha_2}\cdots\frac{L}{D-\alpha_n}X$ operator from right to left.

M-II: The Operator $\frac{1}{f(D)}$ may be decomposed into its partial fractions, then

$$PI = \left[\frac{A_1}{D-\alpha_1}+\frac{A_2}{D-\alpha_2}+\ldots\frac{A_n}{D-\alpha_n}\right]X$$

Solve: $(D^2+a^2)y = \sec ax$

Short Method of finding PI

(1) When $X = e^{ax}$

(i) $\dfrac{L}{f(D)}e^{ax} = \dfrac{L}{f(a)}e^{ax}\,(f(a) \neq 0)$

(ii) $\dfrac{1}{(D-a)^r\phi(D)}e^{ax} = \dfrac{1}{\phi(a)}\dfrac{x^r}{r!}e^{ax}$

Note: $\dfrac{1}{D^n}(1) = \dfrac{x^n}{n!}$

(2) When $X = \sin ax$ or $\cos ax$

Case I: when $f(D)$ can be expressed as $\phi(D^2)$ and $\phi(-a^2) \neq 0$

then $\dfrac{1}{f(D)}\sin ax = \dfrac{1}{\phi(D^2)}\sin ax = \dfrac{1}{\phi(-a^2)}\sin ax$

Similarly, $\dfrac{L}{\phi(D^2)}\cos ax = \dfrac{1}{\phi(-a^2)}\cos ax$

Note : If $f(D)$ conditions odd powers too, then $f(D) = f_1(D^2) + Df_2(D^2)$

$$PI = \frac{1}{f_1(D^2)+Df_2(D^2)}\sin ax = \frac{1}{f_1(-a^2)+Df_2(-a^2)}\sin ax$$

Case II: $\dfrac{1}{D^2+a^2}\sin ax = \dfrac{-x}{2a}\cos ax$ & $\dfrac{1}{D^2+a^2}\cos ax = \dfrac{x}{2a}\sin ax$

(3) When $X = x^m$

$\dfrac{L}{f(D)}x^m =$ use of binomial expansion.

(4) When $X = e^{ax}V$

$\dfrac{l}{f(D)}e^{ax}V = e^{ax}\dfrac{1}{f(D+a)}V$

(5) When $X = xV(x)$

$\dfrac{1}{f(D)}xV(x) = x\dfrac{1}{F(D)}V + \left(\dfrac{d}{dD}\dfrac{1}{f(D)}\right)V$

Q. $(D^2 - 3D + 2)y = e^{5x}$

Ans. $c_1 e^{x} + c_2 e^{2x} + (1/12) e^{5x}$

Q. $D^2 (D+1)^2 (D^2 + D + 1)^2 y = e^{x}$

Q. $(D^2+1)y=\cos 2x$

Q. Solve $(D^2-3D+2)y=\text{Sin}\,3x$

Q. $(D^3+8)y = x^4+2x+1$

Q. $(D^2-2D+1)y = x^2e^{3x}$

Q. $(D^2 - 2D + 1)y = x \sin x$

Homogeneous Linear Equations
or
Cauchy Euler Equations

Form : $(a_0x^nD^n + a_1 x^{n-1}D^{n-1} + ... + xD + a_n)y = X$

eg. $(x^3D^3 + 3x^2D^2 - 2xD + 2)y = 0$

Note : Index of x and the order of derivative is same in each term of such equations.

Working Rule : $(a_0x^nD^n + a_1 x^{n-1}D^{n-1} + ... + a_n)y = X$

(1) Put $x = e^z$ or $z = \ln x$.

(2) Assume $D_1 = d/dz$ and $D = d/dx$,then we have $xD = D_1$, $x^2D^2 = D_1(D_1 - 1)$, $x^3D^3 = D_1(D_1 - 1)(D_1 - 2)$ and so on. equation (1) reduces to $f(D_1)\,y = Z$.

(3) Solve using previously discussed method as $f(D_1)$ has const. coefficient and Z is function of Z only.

Q. Solve $(x^3D^3 + 3x^2D^2 - 2xD + 2)y = 0$

$xD = D_1 \qquad x^2D^2 = D_1(D_1 - 1) \qquad x^3D^3 = D_1(D_1 - 1)(D_1 - 2)$

Put all the values : $[D_1(D_1-1)(D_1-2)+3D_1(D_1-1)-2D_1+2]y=0$

Auxiliary equation roots : 1, 1, -2

$\therefore\ y_c=(c_1+c_2z)e^{z}+c_3e^{-2z}$

Finding PI in case of EC Equation

General Method.

$$\frac{1}{D_1+a}X = x^{-a}\int x^{a-1}X\,dx \text{ where X is a function of } x.$$

Q. $(x^2D^2+3xD+1)y=\dfrac{1}{(1-x)^2}$

Equations Reducible to EC form (Legendre's Linear Equation)

Form : $[a_0(a+bx)^n D^n + a_1(a+bx)^{n-1} D^{n-1} + a_2(a+bx)^{n-2} D^{n-2} + \ldots + a_{n-1}(a+bx)D + a_n]\, y = X$

[Here we have $a + bx$ in place of x]

Replace $a+bx=e^z \Rightarrow z=ln(a+bx)$

$$\frac{dy}{dx}=\frac{dy}{dz}\frac{dz}{dx}=\frac{b}{a+bx}.\frac{dy}{dz}$$

$$\Rightarrow\ (a+bx)\frac{dy}{dx}=b\frac{dy}{dz}$$

$$\Rightarrow\ (a+bx)D=bD_1$$

$\downarrow$

Similarly

$(a+bx)^2D^2=b^2\,D_1(D_1-1)$

$(a+bx)^3D^3=b^3\,D_1(D_1-1)(D_1-2)$ and so on.

Q. **Solve:** $(1+x)^2\,{d^2y}/{dx^2}+(1+x)({dy}/{dx})+y=4\cos\log(1+x)$

Second order Linear Equation with variable Coefficients

Standard form of linear 2nd Order equation $\frac{d^2y}{dx^2}+P\left(\frac{dy}{dx}\right)+Qy=R$ where P, Q, R are functions of x or constants.

Methods

(1) EC method	(2) One known integral	(3) Reduction to Normal form	(4) Change of Independent variable

(5) Variation of parameters.

(2) **When one Integral belonging to the CF is known.**

Given $\frac{d^2y}{dx^2}+P\frac{dy}{dx}+Qy+R$. Let u be known Integral of the complementary function.

i.e $\frac{d^2u}{dx^2}+P\frac{du}{dx}+Qu=0$

↓

Let complete solution of (1) be $y=uv$

↓

$$\frac{d^2V}{dx^2}+\left(P+\frac{2}{u}\frac{du}{dx}\right)\frac{dv}{dx}=\frac{R}{u}$$

↓

Take $\frac{dv}{dx}=q$

↓

$$\frac{dq}{dx}+\left(P+\frac{2}{u}\frac{du}{dx}\right)q=\frac{R}{u}$$

This is linear in q

↓

Solve for q , entry of one const C,

↓

$q = \dfrac{dv}{dx}$; again solve for v. c_2 will make entry.

$\downarrow$

Complete solution: $\boxed{y = uu}$

Note: This will have two arbitrary const.

How to Find u

(1) By Guess

	Candidates	Conditions
	1. e^x	$1 + P + Q = 0$
	2. e^{-x}	$1 - P + Q = 0$
	3. e^{ax}	$a^2 + aP + Q = 0$
	4. x	$P + Qx = 0$
	5. x^2	$2 + 2Px + Qx^2 = 0$
	6. x^m	$m(m-1) + Pmx + Qx^2 = 0$

Q. Solve $xy'' - (2x-1)y' + (x-1)y = 0$ (Solve by reducing the order).

Q. Solve $(x\sin x+\cos x)y''-x\cos x\,y'+y\cos x=0$

Q. Solve $x^2 y_2 - 2x(1+x)y_1 + 2(1+x)y = x^3$

(3) Reduction to normal form [Removal of First Derivative]

Form : $y'' + Py' + Qy = R$

$\downarrow$

Choose $u = e^{-\frac{1}{2}\int p\,dx}$

$\downarrow$

Assume Complete solution as $y = UV$

$\downarrow$

Given equation become $\dfrac{d^2V}{dx^2} + \left(Q - \dfrac{p^2}{4} - \dfrac{1}{2}\dfrac{dP}{dx}\right)V = \dfrac{R}{U}$

Solve for V and final ans is $y = UV$

Q. Solve $y'' - 2\tan x\, y' + 5y = \sec x\, e^{x}$

(4) **Changing the independent variable**

Form: $y'' + Py' + Qy = R$

$$\downarrow$$

$$z = f(x)$$

$$\downarrow$$

$$\frac{d^2y}{dz^2} + \underbrace{\left[\frac{\frac{d^2z}{dx^2} + \frac{Pdz}{dx}}{\left(\frac{dz}{dx}\right)^2}\right]}_{P_1}\frac{dy}{dx} + \underbrace{\frac{Q}{\left(\frac{dz}{dx}\right)^2}}_{Q_1} y = \frac{R}{\left(\frac{dz}{dx}\right)^2}$$

Choose z such that P_1 and Q_1 becomes constant.

$$\downarrow$$

Solve $\frac{d^2y}{dz^2} + P_1\frac{dy}{dx} + Q_1 y = R_1$ by usual methods.

Q. Solve $y'' - y'\cot x - y\sin^2 x = \cos x - \cos^3 x$

(5) **Method of variation of parameters**

Form: $y_2 + Py_1 + Qy = R$ ...(i)

$\downarrow$

Solve: $y_2 + Py_1 + Qy = 0$

$\downarrow$

General solution of (1) $y = CF + PI$

where C.F. $= c_1u + c_2v$

and PI $= uf(x) + vg(x)$

where $f(x) = -\int \frac{vR}{W} dx$

and $g(x) = \int \frac{uR}{W} dx$

where W = wronskian of u & v

$$\begin{vmatrix} u & v \\ u_1 & v_1 \end{vmatrix} = uv_1 - u_1v$$

Q. Solve $x\frac{dy}{dx} - y = (x-1)\left(\frac{d^2y}{dx^2} - x + 1\right)$ using method of variation of parameters.

Laplace Transform

The Laplace transforms of $f(t)$, *denotes* by $L(f(t))$ or $F(s)$ is defined by

$F(s) = Lf(t) = \int_0^{\infty} e^{-st} f(t)dt$, where s is a parameter which may be real or complex. $f(t)$ is called inverse Laplace transform of F(s).

Advantages

- No need to find the solution of homogeneous DE first to solve the DE.
- Reduce the problem to mere algebraic manipulations.
- Directly incorporate the initial condition in the solution.

Transforms of Elementary Functions

$f(t)$	$f(s)$
1	$1/s$
t^n	$\frac{\lceil n+1}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\sinh at$	$\frac{a}{s^2-a^2}$
$\cosh at$	$\frac{s}{s^2-a^2}$
$\sin at$	$\frac{a}{s^2+a^2}$
$\cos at$	$\frac{s}{s^2+a^2}$

Q. Find the laplace transform of t^n $L(t^n) = \int_0^{\infty} e^{-st} t^n \, dt$

Q. Find the laplace transform of $\cos at$.

Q. Find the laplace transform of $\sin\sqrt{t}$

Theorems/Properties

(1) $L\,f(t)=F(s) \rightarrow L\,e^{at}f(t)=F(s-a)$: First translation theorem.

(2) $Lf(t-a)=e^{-as}F(s)$: Second translation Theorem.

(3) $Lf(at)=\frac{1}{a}F(s/a)$: Change of scale property.

(4) Laplace transform of n^{th} derivative.

$$Lf^{n}(t)=s^{n}F(s)-s^{n-1}f(0)-s^{n-2}f'(0)\ldots-f^{n-1}(0)$$

(5) $$Lt^{n}f(t)=(-1)^{n}\frac{d^{n}F(s)}{ds^{n}}$$

(6) $$L\frac{f(t)}{t}={}_{s}\int^{\infty}F(s)ds$$

(7) $$L\int_{0}^{t}\frac{f(t)}{t}=\frac{1}{s}F(s)$$

(8) Transform of periodic function : $f(t)=f(t+T)$

$$F(s)=\frac{1}{1-e^{-sT}}\int_{0}^{T}e^{-st}f(t)dt$$

(9) $$\int_{0}^{\infty}\frac{f(t)}{t}dt=\int_{0}^{\infty}F(s)ds$$

(10) Initial value Theorem

$$\lim_{t\to 0}f(t)=\lim_{s\to\infty}sF(s)$$

(11) Final value Theorem

$$\lim_{t\to\infty}f(t)=\lim_{s\to 0}sF(s)$$

(12) $\int_{0}^{\infty}e^{-at}f(t)dt=F(a)$ (Used for evaluating Integrals)

(13) $L\{\delta(t)\}=1$

The Inverse Laplace Transform

If $F(s)$ is laplace transform of $f(t)$, then $f(t)$ is called inverse laplace transform of F(s).

e.g $L(e^{at}) = \dfrac{1}{(s-a)} \Rightarrow L^{-1}\left(\dfrac{1}{s-a}\right) = e^{at}$ Laplace inverse of some basic functions :

$F(s)$	$f(t)$
$\dfrac{1}{s}$	1
$\dfrac{1}{s^{n+1}}$	$\dfrac{t^n}{\left\lvert\overline{n+1}\right.}$
$\dfrac{1}{s-a}$	e^{at}
$\dfrac{1}{s^2+a^2}$	$\dfrac{\sin at}{a}$
$\dfrac{s}{s^2+a^2}$	$\cos at$

Q. $\frac{1}{\sqrt{s}}$

Q. $\frac{(3s-16)}{(s^2-16)}$

Convolution

Let $f(t)$ and $g(t)$ be two funtions , then the convolution of them is denoted by .

$$\boxed{f * g = \int_0^t f(u).g(t-u)\,du}$$

The Convolution theorem or convolution Property

If $L^{-1}F(s) = f(t)$ and $L^{-1}(G(S)) = g(t)$

$$\boxed{L^{-1}(F(s).G(s) = \int_0^t f(u).g(t-u)du}$$

Q. Use the convolution theorem to evaluate.

(i) $L^{-1}\left\{\frac{s^2}{(s^2+a^2)^2}\right\}$ (ii) $L^{-1}\left\{\frac{1}{(s-1)^5(s+2)}\right\}$

Solution Of ODE with Const. Coefficient

$(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \ldots + a_n)y = f(t)$

Subject to initial conditions

$y(0) = r_0$, $y'(0) = r_1$, $y''(0) = r_2, \ldots.$

Take Laplace transform on both sides and solve.

Q. Solve $(D^2 + 1)y = 6\cos 2t$. If $y = 3$, $Dy = 1$, when $t = 0$.

NON CONSTANT COEFFICIENTS IVP